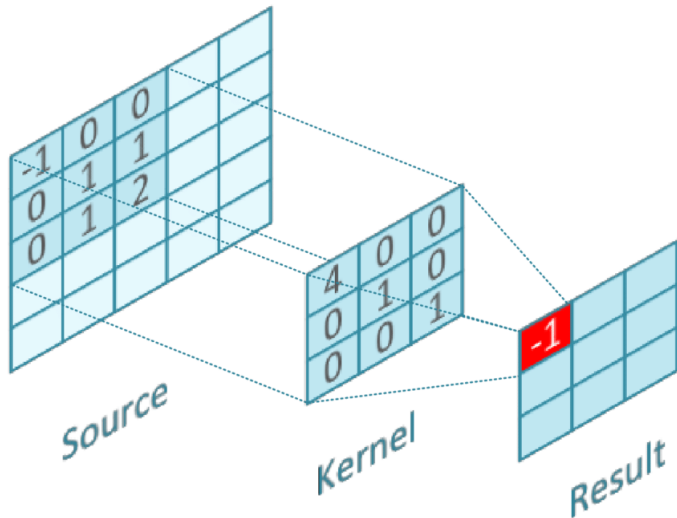


Deep Learning

Vazgen Mikayelyan

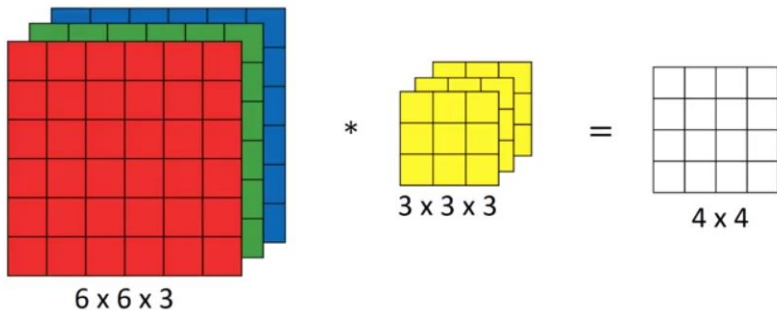


2D Convolution

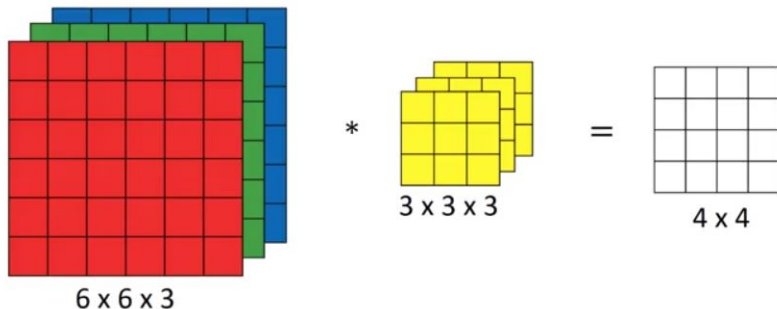


Convolution on RGB Images

Convolution on RGB Images

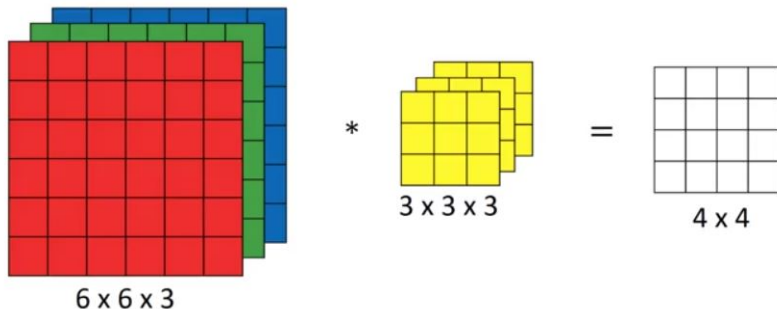


Convolution on RGB Images



- How many dimensions has convolution kernel in general?

Convolution on RGB Images



- How many dimensions has convolution kernel in general?
- What does 1×1 convolution do?

Valid and Same Convolution

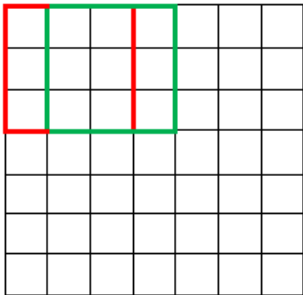
- Padding = Same: means the input image ought to have zero padding so that the output in convolution doesn't differ in size as input.

Valid and Same Convolution

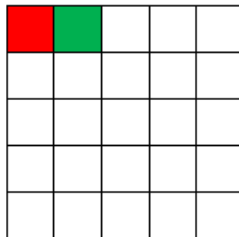
- Padding = Same: means the input image ought to have zero padding so that the output in convolution doesn't differ in size as input.
- Padding = Valid: means we don't add the zero pixel padding around the input matrix, and its like saying, we are ready to loose some information.

Strided Convolution

7 x 7 Input Volume

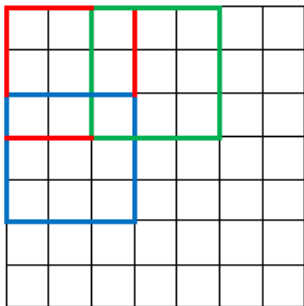


5 x 5 Output Volume

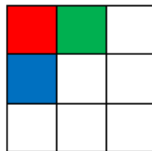


Strided Convolution

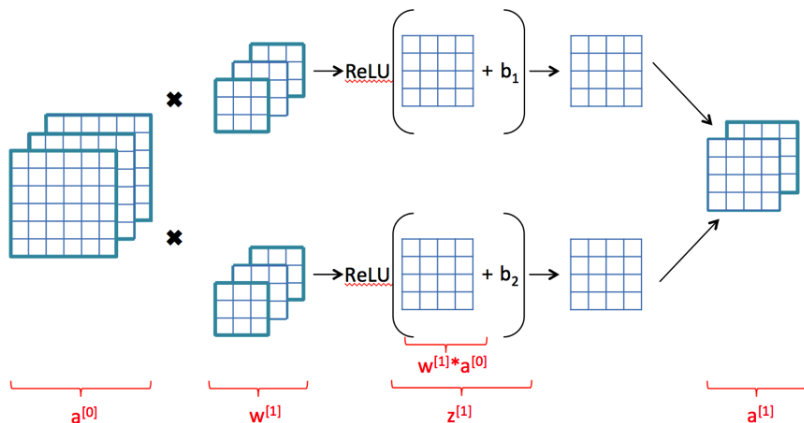
7 x 7 Input Volume



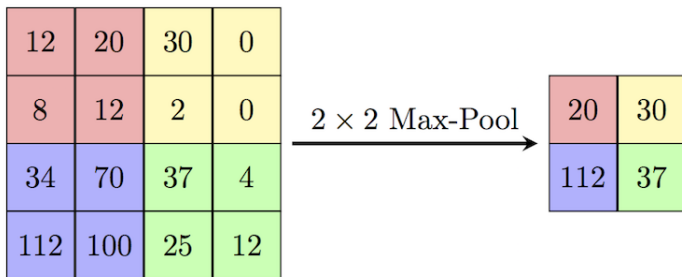
3 x 3 Output Volume



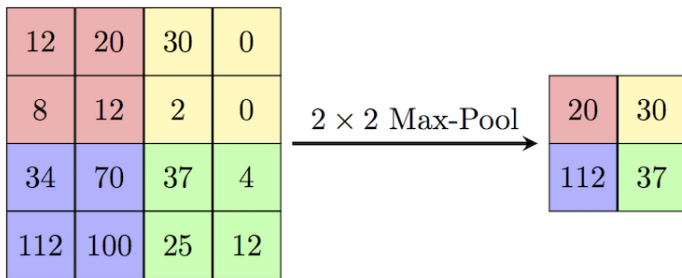
One Layer of CNN



Pooling Layers

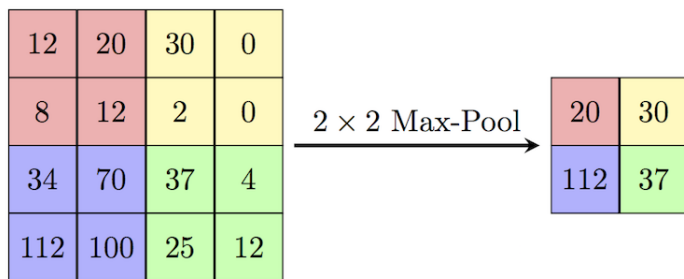


Pooling Layers



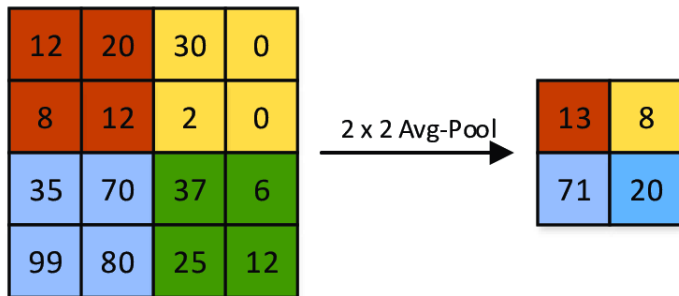
- Number of channels is the same after pooling layer.

Pooling Layers



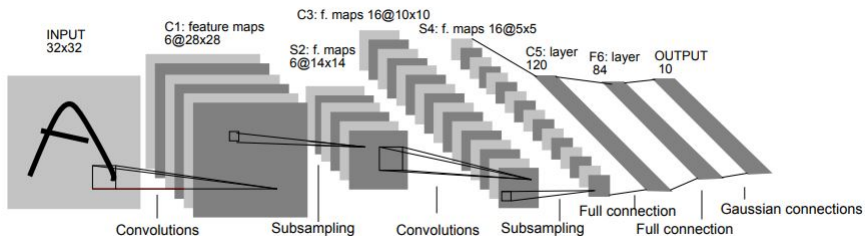
- Number of channels is the same after pooling layer.
- There are not trainable parameters in this layer.

Pooling Layers

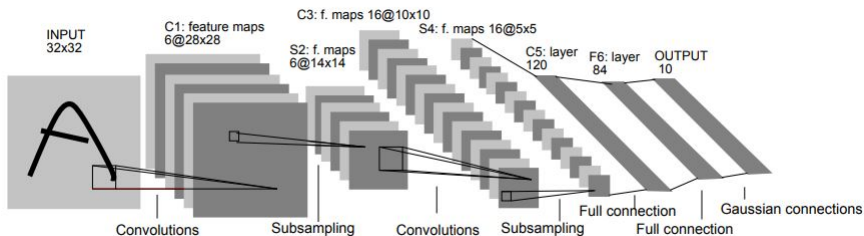


1 Famous CNNs

LeNet-5 (1998)

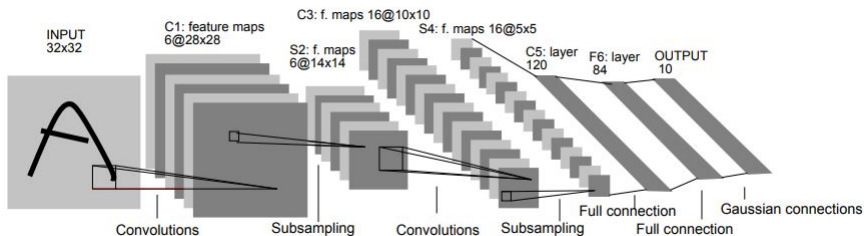


LeNet-5 (1998)



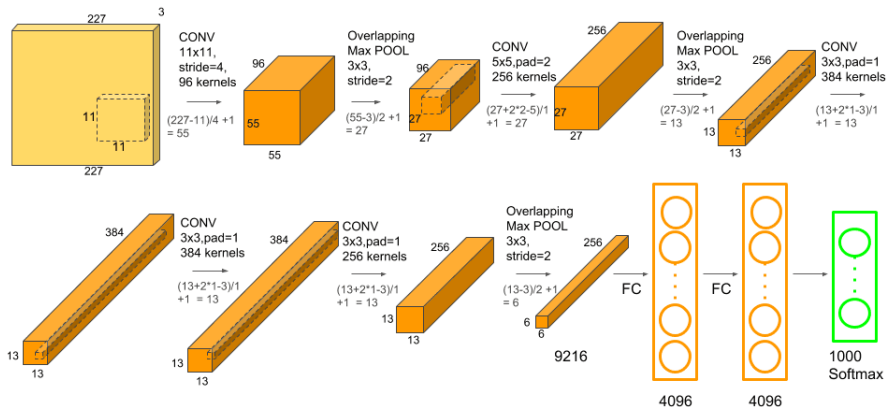
- Activation functions are sigmoids and hyperbolic tangents.

LeNet-5 (1998)



- Activation functions are sigmoids and hyperbolic tangents.
- LeNet-5 has approximately 60k parameters.

AlexNet (2012)



- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- Local response normalization?

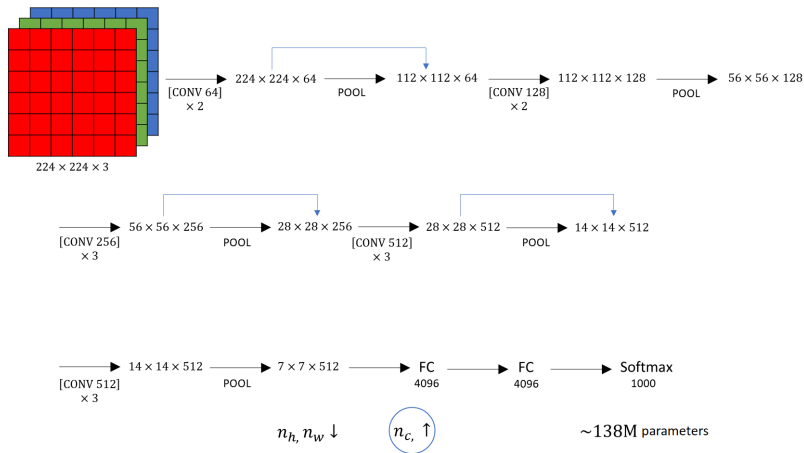
- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- Local response normalization?
- AlexNet has approximately 60M parameters.

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- Local response normalization?
- AlexNet has approximately 60M parameters.
- Accuracies on ImageNet: Top1=63.3%, Top5=84.6%.

VGG-16 (2014)

CONV = 3×3 filter, $s=1$, same

MAX-POOL = 2×2 , $s=2$



- Fixed kernel size.

- Fixed kernel size.
- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.

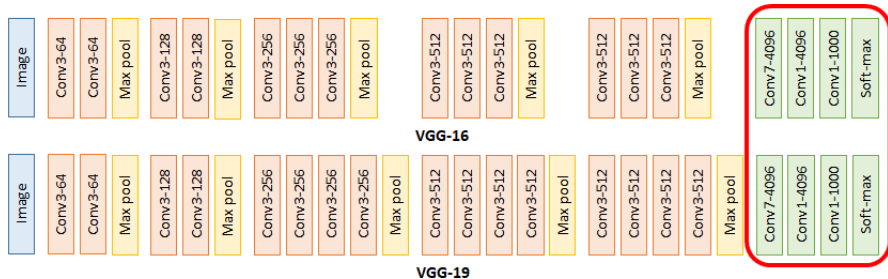
VGG-16 (2014)

- Fixed kernel size.
- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- VGG-16 has approximately 138M parameters.

VGG-16 (2014)

- Fixed kernel size.
- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- VGG-16 has approximately 138M parameters.
- Accuracies on ImageNet: Top1=74.4%, Top5=91.9%.

VGG-19 (2014)

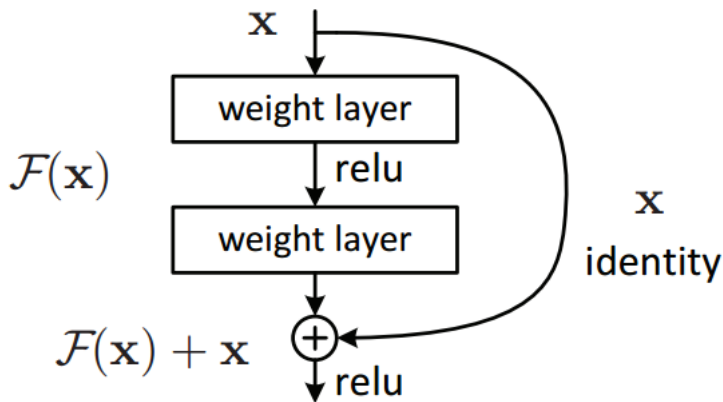


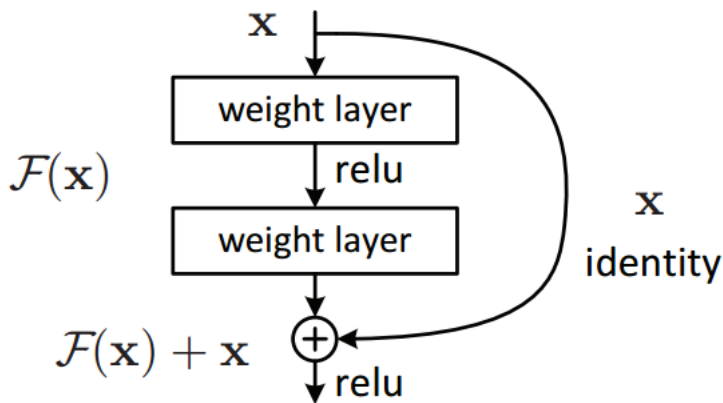
- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- VGG-19 has approximately 144M parameters.

VGG-19 (2014)

- The ReLU non-linearity is applied to the output of every convolutional and fully-connected layer.
- VGG-19 has approximately 144M parameters.
- Accuracies on ImageNet: Top1=74.5%, Top5=92%.





Identity function is easy to learn for residual block.